

Sealed Experiment Record

Sealed 2026-10-08 · Signed by the K-Veritas server
Session 76a53d82 · Data hash d74a...00ab

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-08 06:18 and 08:41 UTC. Reported results: final_val_loss 1.1907, c_inference_tokens_per_second 69.4935. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	76a53d82-f82f-457a-affd-094ac40d4683
Initialized	2026-10-08T06:18:17Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	dd2958b8e7cab7af968712377075cf14ba515dfbffc84f7f7a242dd21a4df7e3
Disclosure	open
Source code, run 1	8303b2b60f2af89c94bc42cd3b61b81af9bc777acffe258aa9e24f33a75a1877

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	2 hours 23 minutes	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
train_loss	1.1897	14000 (8 readings)	explicit
val_loss	1.1907	14000 (8 readings)	explicit
c_inference_tokens_per_second	69.4935		explicit

Inline claim	Value	Phase	Output line
final_val_loss	1.1907		29
c_inference_tokens_per_second	69.4935		39

Seed committed: seed:1337, output line 1, 2026-10-08T06:18:35Z.

4 Execution evidence

Coherence

Coherent (0.50): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	8580.0 s	8.46 GB	99%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 66 tracked files, 4 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	06:18:35	d7713c300aa49862	66
3	run_end	08:41:48	d7713c300aa49862	0

2 snapshots with no file change omitted here; full timeline in Appendix B.

Withheld by .kveritasignore	Size	Hash
data/TinyStories_all_data/data00.json	<1GB	f2d7724a43d4ec5c936f83fa
data/TinyStories_all_data/data01.json	<1GB	62f22c99d153ce9430ab9776
data/TinyStories_all_data/data02.json	<1GB	bd46a31dbc2c2b09854a8185
data/TinyStories_all_data/data03.json	<1GB	256ea4343c10c714b8e8115e
data/TinyStories_all_data/data04.json	<1GB	ccd6d7e4318c5d81eed55cb6
data/TinyStories_all_data/data05.json	<1GB	4a8b1ed8a9600e92da3ace36
data/TinyStories_all_data/data06.json	<1GB	691d495b059b33e89dc8cc50
data/TinyStories_all_data/data07.json	<1GB	9289d8fec5ac2a9fd98291ae
data/TinyStories_all_data/data08.json	<1GB	63a38566b660d341f038af7f
data/TinyStories_all_data/data09.json	<1GB	a65da4d48eba93a5b313c2ea
data/TinyStories_all_data/data10.json	<1GB	3c8ea65dd88db0203925e81b
data/TinyStories_all_data/data11.json	<1GB	02f740a3e2d80099bf24fbf5
data/TinyStories_all_data/data12.json	<1GB	e68006a1fcb63585ee4f3c46
data/TinyStories_all_data/data13.json	<1GB	6af2f7ad317646616e9b7516
data/TinyStories_all_data/data14.json	<1GB	098674bb4cebd07359c1731e
data/TinyStories_all_data/data15.json	<1GB	7c211721245c32c2e0358d81
data/TinyStories_all_data/data16.json	<1GB	a48214b2d35367c776d47aca
data/TinyStories_all_data/data17.json	<1GB	0f59647f2d6e3136116283c6
data/TinyStories_all_data/data18.json	<1GB	09194081a607d84e37d75f91
data/TinyStories_all_data/data19.json	<1GB	03248b8edadd89fe246d2cf3
data/TinyStories_all_data/data20.json	<1GB	66c3ccee11ec2535931b53d7
data/TinyStories_all_data/data21.json	<1GB	901ff607a5feda7fe58855d2
data/TinyStories_all_data/data22.json	<1GB	ca24443827b3396b7887e59b
data/TinyStories_all_data/data23.json	<1GB	f5c9eaa4bf31accc5c47beaa
data/TinyStories_all_data/data24.json	<1GB	4d1a9b0c91be252e0b8f3595

Withheld by .kveritasignore	Size	Hash
data/TinyStories_all_data/data25.json	<1GB	55109751864aab952cfe4260
data/TinyStories_all_data/data26.json	<1GB	0497670618e96ba3e2e3134f
data/TinyStories_all_data/data27.json	<1GB	ccee5fa376e342e0cb946520
data/TinyStories_all_data/data28.json	<1GB	03ca389f0d3f824a3e136cc8
data/TinyStories_all_data/data29.json	<1GB	80d58544568eb80e17431fd4
data/TinyStories_all_data/data30.json	<1GB	4acf0701831f3d0fe0721ecc
data/TinyStories_all_data/data31.json	<1GB	c9dfec744d4e38dd6b4e0f16
data/TinyStories_all_data/data32.json	<1GB	5e6d1d2cbd8b59ef81a5c920
data/TinyStories_all_data/data33.json	<1GB	0f66eee2b434e51d81a3c855
data/TinyStories_all_data/data34.json	<1GB	3aad7c557e2f89554982b2d
data/TinyStories_all_data/data35.json	<1GB	59eda54b42394fb77a3f99a0
data/TinyStories_all_data/data36.json	<1GB	ccae069b8d47e679cc6340b1
data/TinyStories_all_data/data37.json	<1GB	a949f6e6e7a638083d12bb2a
data/TinyStories_all_data/data38.json	<1GB	a00429f44b9524ee89c4895e
data/TinyStories_all_data/data39.json	<1GB	d760e35067c22f04aed9e643
data/TinyStories_all_data/data40.json	<1GB	f113b0cc5b165b00d098bfc4
data/TinyStories_all_data/data41.json	<1GB	b3e7588cba010d2ff292535a
data/TinyStories_all_data/data42.json	<1GB	72666d11682d8429b61df1da
data/TinyStories_all_data/data43.json	<1GB	318af5f4789afa2dec6c7528
data/TinyStories_all_data/data44.json	<1GB	1382ae113fd8b1b6ffad27d8
data/TinyStories_all_data/data45.json	<1GB	ba014bd1567f6dd7bf993bf9
data/TinyStories_all_data/data46.json	<1GB	86526c465d99b368a1d7eba2
data/TinyStories_all_data/data47.json	<1GB	0d3788b58656bd614935abb0
data/TinyStories_all_data/data48.json	<1GB	e618850771d1eb5a29e75775
data/TinyStories_all_data/data49.json	<1GB	d5aa09aa7c90e7c337c4ce87

Activity: 128 files read, 0 files written, 24 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-08T08:42:07.323617+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	d74a38cd96e3053227f568c286dcfa5f7a8d639d64e95e1b7cf8b32a2b5f00ab
Code bundle hash	ea76c39ccd4d04686ac8a8d77aa09a8083de9812b35d1d3af6ca7b97d6db8cd3

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
train_loss	10.4169	0	13	explicit
val_loss	10.4173	0	14	explicit
train_loss	1.4774	2000	15	explicit
val_loss	1.4786	2000	16	explicit
train_loss	1.3377	4000	17	explicit
val_loss	1.3388	4000	18	explicit
train_loss	1.2822	6000	19	explicit
val_loss	1.2827	6000	20	explicit
train_loss	1.2469	8000	21	explicit
val_loss	1.2484	8000	22	explicit
train_loss	1.2192	10000	23	explicit
val_loss	1.2203	10000	24	explicit
train_loss	1.1997	12000	25	explicit
val_loss	1.2006	12000	26	explicit
train_loss	1.1897	14000	27	explicit
val_loss	1.1907	14000	28	explicit
c_inference_tokens_per_second	69.4935		38	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	2	06:18:36	23.8 s	8.46 GB	99%, 8642 MB
sample	30	08:41:11	0.1 s	0.07 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	06:18:35	d7713c300aa49862	66
1	phase train	06:18:55	d7713c300aa49862	0
2	phase sample	08:41:30	d7713c300aa49862	0
3	run_end	08:41:48	d7713c300aa49862	0

Event	Path or command	Hash
read	kv_run.sh	
read	train.py	
read	.kveritasignore	
read	.github/workflows/build.yml	
read	configurator.py	
read	data/TinyStories_all_data/data00.json	
read	data/TinyStories_all_data/data01.json	
read	data/TinyStories_all_data/data02.json	
read	model.py	
read	data/TinyStories_all_data/data03.json	
read	data/TinyStories_all_data/data04.json	

Event	Path or command	Hash
read	data/TinyStories_all_data/data05.json	
read	data/TinyStories_all_data/data06.json	
read	data/TinyStories_all_data/data07.json	
read	data/TinyStories_all_data/data08.json	
read	data/TinyStories_all_data/data26.bin	
read	data/TinyStories_all_data/data09.json	
read	data/TinyStories_all_data/data10.json	
read	data/TinyStories_all_data/data11.json	
read	data/TinyStories_all_data/data12.json	
read	data/TinyStories_all_data/data13.json	
read	data/TinyStories_all_data/data14.json	
read	data/TinyStories_all_data/data15.json	
read	data/TinyStories_all_data/data16.json	
read	data/TinyStories_all_data/data17.json	
read	data/TinyStories_all_data/data18.json	
read	data/TinyStories_all_data/data19.json	
read	data/TinyStories_all_data/data20.json	
read	data/TinyStories_all_data/data21.json	
read	data/TinyStories_all_data/data22.json	
read	data/TinyStories_all_data/data23.json	
read	data/TinyStories_all_data/data24.json	
read	data/TinyStories_all_data/data25.json	
read	data/TinyStories_all_data/data26.json	
read	data/TinyStories_all_data/data00.bin	
read	data/TinyStories_all_data/data27.json	
read	data/TinyStories_all_data/data28.json	
read	data/TinyStories_all_data/data29.json	
read	data/TinyStories_all_data/data30.json	
read	data/TinyStories_all_data/data31.json	
read	data/TinyStories_all_data/data32.json	
read	data/TinyStories_all_data/data33.json	
read	data/TinyStories_all_data/data34.json	
read	data/TinyStories_all_data/data35.json	
read	data/TinyStories_all_data/data36.json	
read	data/TinyStories_all_data/data37.json	
read	data/TinyStories_all_data/data38.json	
read	data/TinyStories_all_data/data39.json	
read	data/TinyStories_all_data/data40.json	
read	data/TinyStories_all_data/data41.json	
read	data/TinyStories_all_data/data42.json	
read	data/TinyStories_all_data/data43.json	
read	data/TinyStories_all_data/data44.json	
read	data/TinyStories_all_data/data45.json	
read	data/TinyStories_all_data/data46.json	
read	data/TinyStories_all_data/data47.json	
read	data/TinyStories_all_data/data48.json	
read	data/TinyStories_all_data/data49.json	
read	export.py	
read	run.c	
read	run.ipynb	

Event	Path or command	Hash
read	runq.c	
read	sample.py	
read	test.c	
read	test_all.py	
read	tinystories.py	
read	tokenizer.py	
read	win.c	
read	win.h	
read	LICENSE	
read	Makefile	
read	README.md	
read	assets/llama_cute.jpg	
read	build_msvc.bat	
read	doc/stories260K.md	
read	doc/train_llama_tokenizer.md	
read	requirements.txt	
read	tokenizer.bin	
read	tokenizer.model	
read	data/TinyStories_all_data/data24.bin	
read	data/TinyStories_all_data/data20.bin	
read	data/TinyStories_all_data/data12.bin	
read	data/TinyStories_all_data/data05.bin	
read	data/TinyStories_all_data/data45.bin	
read	data/TinyStories_all_data/data27.bin	
read	data/TinyStories_all_data/data10.bin	
read	data/TinyStories_all_data/data30.bin	
read	data/TinyStories_all_data/data17.bin	
read	data/TinyStories_all_data/data32.bin	
read	data/TinyStories_all_data/data22.bin	
read	data/TinyStories_all_data/data13.bin	
read	data/TinyStories_all_data/data04.bin	
read	data/TinyStories_all_data/data37.bin	
read	data/TinyStories_all_data/data39.bin	
read	data/TinyStories_all_data/data11.bin	
read	data/TinyStories_all_data/data25.bin	
read	data/TinyStories_all_data/data49.bin	
read	data/TinyStories_all_data/data01.bin	
read	data/TinyStories_all_data/data43.bin	
read	data/TinyStories_all_data/data19.bin	
read	data/TinyStories_all_data/data33.bin	
read	data/TinyStories_all_data/data48.bin	
read	data/TinyStories_all_data/data34.bin	
read	data/TinyStories_all_data/data31.bin	
read	data/TinyStories_all_data/data29.bin	
read	data/TinyStories_all_data/data21.bin	
read	data/TinyStories_all_data/data23.bin	
read	data/TinyStories_all_data/data42.bin	
read	data/TinyStories_all_data/data46.bin	
read	data/TinyStories_all_data/data36.bin	
read	data/TinyStories_all_data/data44.bin	

Event	Path or command	Hash
read	data/TinyStories_all_data/data14.bin	
read	data/TinyStories_all_data/data40.bin	
read	data/TinyStories_all_data/data47.bin	
read	data/TinyStories_all_data/data03.bin	
read	data/TinyStories_all_data/data28.bin	
read	data/TinyStories_all_data/data38.bin	
read	data/TinyStories_all_data/data06.bin	
read	data/TinyStories_all_data/data35.bin	
read	data/TinyStories_all_data/data07.bin	
read	data/TinyStories_all_data/data09.bin	
read	data/TinyStories_all_data/data15.bin	
read	data/TinyStories_all_data/data16.bin	
read	data/TinyStories_all_data/data18.bin	
read	data/TinyStories_all_data/data02.bin	
read	data/TinyStories_all_data/data08.bin	
read	data/TinyStories_all_data/data41.bin	
read	run	
process	python -u train.py --dim=288 --n_layers=6 --n_heads=6 --max_iters=15000 --...	
process	grep -E ^step tokens per iteration ^[0-9]+0000 \	
process	tee ./train.out	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	pid 1065472	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/025-karpathy-llama2.c/env/bin/python /var/tmp/kv-repro/0...	
process	pid 1353625	
process	./run /var/tmp/kv-repro/025-karpathy-llama2.c/data/run/model.bin -t 0.8 -n 25...	
process	tee ./sample.out	

C Run history

Invocation 1, OK, 2 hours 23 minutes, 39 output lines, started 2026-10-08T06:18:17

```
metric_hash e6b3be36a7991ac6bc57f4d15c27d68daa684b0f349fe0de4943abce1d2ee1e1
run_digest 144b6687fa7acf364125ec391d173a4ed6ec4f4ec86b6b8b22ad26bed5d923ce
anchored    at run end
```

D Raw seal fields and hashes

Seal

```
data_hash          d74a38cd96e3053227f568c286dcfa5f7a8d639d64e95e1b7cf8b32a2b5f00ab
signed_message_hash 315d7e49519bbcc281193a0aa4cc96a95929b5c8f6653d14ef9ec54fcc72bb20
nonce             b942a2ef01ca707654aeff2c2da70dee
signed_at         2026-10-08T08:42:07.323617+00:00
sealed_at         2026-10-08T08:42:07.387086314Z
checkout_bundle   ea76c39ccd4d04686ac8a8d77aa09a8083de9812b35d1d3af6ca7b97d6db8cd3
```

Signature (base64)

```
Rx0+CHZ2xRBjsr70nEf9aVw4BjD+GIVneK43mXGkxY5dPmky8omreR0nDaUltj0LPgAjpQf8oCplXC lUgbtb0vtw
lYn0dbv6V57P3UmFbbdbBLv lwiFd+1hPGdw0THcnZQ32R4UiIRIaI1UeiDoEJFUqG4Y7dTzqGahG5BZBLw3lqUM
fR/TLp1NdGJWvi9Xh+Dn09g5j/9CTVsRmMH2P/TDq3ve0CpDFvwjhBF5bdrYN5RYnsboJaSAd7xr7l9xa0015Rv5
admJdCrjI l0bIc242zt0ADmzpg+BwETbnCwQnTJg/ZeLDHj0FXMtwyAQzgQvdrc9+Gz0g/fCgg/UQdZ3tKHsFcb/
0NFBa0rnh0BKvnfjiy2G/b9KC5v5vCF11RY164vEPt9QE3+nWR4oYS00g4pSdpZ8WtRf6ZjDzxUKLpTZ/35pCAjt
wqigFjbcugKtUb6ARbd0zfs/9Vq0bBkJLz/ySkoiz6bBcR0kuXB/Aw6tzLTjKX0z0bN03MoNyzsMwonMADkvtjG
PzBzjCDdSH/8Q/EHCS9us4jqb8uLbYQ1q5PWEvh6lVEKYG0eai0oND/hi5/MB0y8va0auQ7U1dqHDQjTE7katzzj
ecVL0Xdqt6sA7MXJ3Kzman0Wp3iqW1Ta0M3vhUD+vtTeq+RT49q2+CkrzfK8wCJqfzI=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgqhkiG9w0BAQEFAAOCAg8AMIICCGKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAT4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNpPxYa5X
7qU95EFEEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNlrvPU9TAqdbtozdhdQ+D
AhfTRuJNgWEkjwD5lboVhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqeS7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPGpZ242NH8JgE6CNF7ToL9syasS67u7UMgICnGeNHnCy9bLgPU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdW3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 3f9dbdf4b1eb3441c70a953e13a0fda3770b36c9cc72b254cfbbcc476596c1f46
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics e6b3be36a7991ac6bc57f4d15c27d68daa684b0f349fe0de4943abce1d2ee1e1
digest 144b6687fa7acf364125ec391d173a4ed6ec4f4ec86b6b8b22ad26bed5d923ce
kv_run.sh (unchanged)
pre 858e346fdaed054b2b0cd029b29e593416d02f95466d9aff406d5cb117248a87
post 858e346fdaed054b2b0cd029b29e593416d02f95466d9aff406d5cb117248a87
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```